

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **Linux Yocto build processes** for embedded automotive systems, including **kernel customization**, **file system patching**, and **container-based build environments** supporting multi-SoC device deployment across 8+ production ECU hardware platforms at HCL Technologies.
- Designed and deployed **vision framework architecture** for TI-based SoC devices, managing **project repository setup** aligned with Valeo standards, **toolchain adaptation**, and **hardware-software interface integration** (GPIOs, SPI, I2C, UART, Ethernet, Memory) reducing time-to-market by 35%.
- Implemented **end-to-end AI solution** for requirement analysis and **code generation** targeting automotive platform deployment, achieving **40% reduction** in manual code authoring and enabling seamless customer project integration via **CMake**, **compiler/linker optimization**, and automated deployment pipelines.
- Executed comprehensive **black-box testing**, **integration testing**, and **unit testing** across firmware and middleware modules; conducted **static code analysis** using **Klocwork** and enforced **MISRA C/C++ coding standards**, achieving **100% code review compliance** and **zero critical defects** in production releases.
- Developed low-level firmware and driver code in **C** and **C++** adhering to **ASPICE** and **ISO 15504** quality processes; maintained comprehensive **software requirement specifications** and **module design documentation** in **Code Beamer configuration management**, ensuring regulatory traceability and audit readiness.
- Managed **wiki-page documentation** and **Google Site design** for platform onboarding and technical reference; coordinated formal **code reviews** and **document reviews** with cross-functional teams; maintained all work products in version control with **Git branching** strategies and **CI/CD Jenkins pipelines** for automated build validation.

SKILLS

Embedded Systems & Low-Level Software – Firmware development, driver development, middleware, hardware-software interface, GPIO control, SPI, I2C, UART, Ethernet, Memory management.

Embedded Linux & Build Systems – Linux Yocto build process, kernel customization, file system patching, container-based build environments, CMake, compiler/linker optimization, toolchain adaptation.

Programming Languages – C, C++, Python, Java, Shell scripting.

Automotive Software Standards & QA – MISRA C/C++ coding standards, ASPICE, ISO 15504, static code analysis (Klocwork), code review processes, testing (black-box, integration, unit).

Development Tools & Version Control – Git, Jenkins CI/CD pipelines, Code Beamer configuration management, wiki-page documentation, Google Site design.

Hardware Platforms & Interfaces – Texas Instruments SoC devices, ECU hardware development, multi-platform porting, vision framework deployment.

Software Development Practices – Requirements analysis, software requirement specification, module design, documentation, formal code reviews, change control.

EDUCATION AND TRAINING

PROJECTS

Automotive Vision Framework SoC Migration Platform (HCL Technologies)

- Architected and implemented **vision framework** porting strategy across three generations of **TI SoC devices**, establishing **Valeo-compliant project repository structure**, standardized **toolchain configuration**, and reproducible **Linux Yocto build environments** containerized with Docker.
- Executed **hardware-software interface integration** spanning **GPIO, SPI, I2C, UART, and Ethernet** protocols; developed low-level **C/C++ firmware drivers** and validated across ECU test benches, achieving **99.8% hardware compatibility** and eliminating **12 critical porting defects**.
- Developed **AI-powered code generation pipeline** for automated requirement-to-code mapping, reducing manual firmware authoring by **40%** and accelerating customer project deployment cycles from **16 weeks to 10 weeks**; integrated with **CMake build automation** and **Jenkins CI/CD** for zero-touch deployment.
- Executed full-cycle **black-box, integration, and unit testing** protocols; performed **static code analysis using Klocwork** across **850K lines of C/C++**, enforced **MISRA standards**, and achieved **100% critical/high-severity defect resolution** prior to production handoff.
- Maintained comprehensive **software requirement specifications** and **module design documentation** in **Code Beamer**; coordinated formal code and document reviews with senior architects; documented all architecture decisions via **wiki-page** and **Google Site** enabling **50% faster onboarding** of new team members.

Embedded Linux Kernel Customization & Build Optimization Initiative (HCL Technologies)

- Designed and operationalized **Linux Yocto custom build framework** with optimized **kernel configuration**, **device tree overlays**, and **file system patches** reducing boot-time latency by **28%** across automotive ECU deployments.
- Established **container-based build environment** using Docker, enabling reproducible cross-compilation and eliminating build-environment drift across **8 development teams**; documented **compiler/linker best practices** and **CMake module standardization** via wiki achieving **95% adoption**.
- Enforced **ASPICE and ISO 15504** compliance across build-system configuration; performed **static analysis** of kernel patches and middleware modules; maintained audit trail in **Code Beamer** satisfying automotive regulatory requirements.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)